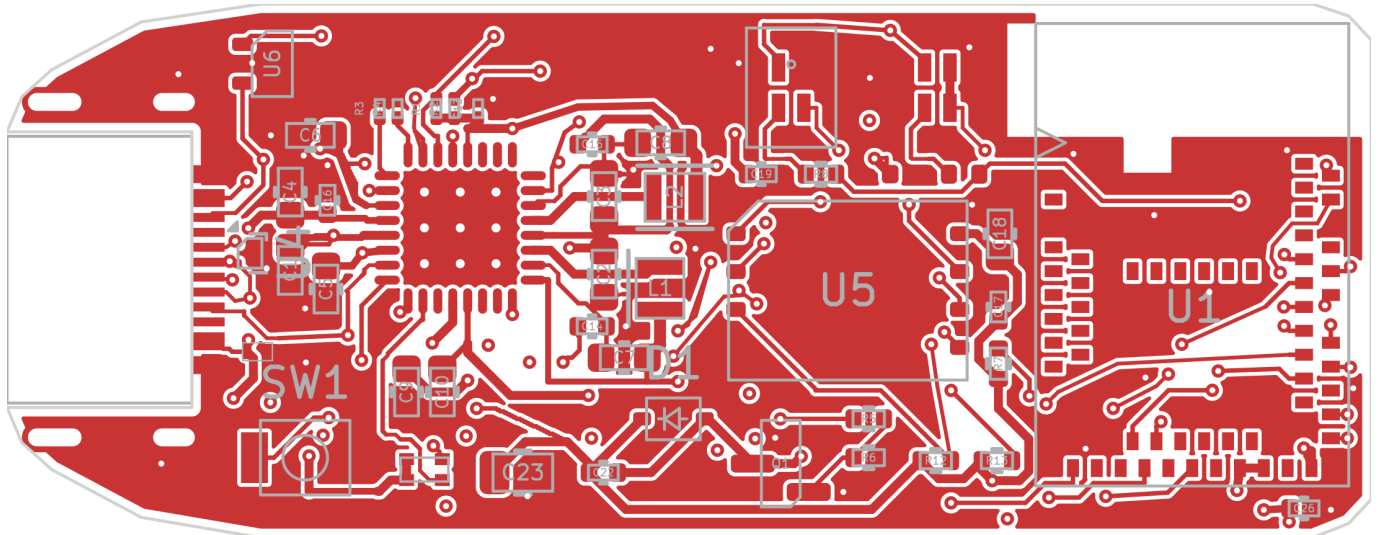


PCB fabrication and assembly release

Corrected prototype manufacturing data. This document identifies the frozen PCB, its required fabrication process and the first-article checks. It is not a finished-wearable qualification certificate.



Actual native front-copper and placement export. USB-C is at the left cutout; the antenna keepout is at the upper right. Not a 1:1 print template.

Released geometry	Verified file checks
45.75 x 17.88 x 0.80 mm; 4 copper layers	DRC 0 / unrouted 0 / schematic parity 0 / ERC 0
47 machine-fitted parts: 44 top, 3 bottom	All 45 named nets pass independent copper connectivity
1,310 track segments / 137 vias	No ignored DRC/ERC categories or individual exclusions

Corrections included

Moved U4 away from C1, repaired its three short front-copper branches and removed their negative courtyard exception. Corrected U2/U4 mask openings. Specified epoxy-filled, copper-capped vias. Fixed the four component selections and three packaging aliases across schematic, board, libraries and BOMs; named the selected Jauch battery consistently.

Use the whole release

Native project: **pcb/electrical/Anticipy_R1_E1_PROTOTYPE.kicad_pro**. Use the matching Gerbers, drills, IPC netlist, placement CSV, fitted BOM, DNP list, paste and via-coordinate CSV in **pcb/manufacturing/**. The supplied STEP shows the board and nominal component envelopes; Gerbers and drills control manufacture.

PCB SHA-256: 78fe28acde6ceb3ab33aac8af503d4c4
18540cce31979cc91d3add5e5d0bec36

Fabrication requirements

Use the exact outline and USB cutout. Do not scale. Target outline tolerance ± 0.10 mm and finished thickness 0.80 ± 0.08 mm. Material: FR-4; ENIG exposed finish, green mask and white silk. This process is more demanding than an ordinary tented-via four-layer order.

Layer, top to bottom	Nominal thickness	Dielectric basis
F.Cu	0.0350 mm	Outer nominal 1 oz
Prepreg	0.0994 mm	3313 RC57%; Dk 4.10
In1.Cu	0.0152 mm	Inner copper
Core	0.5000 mm	NanYa NP-155F; Dk 4.48
In2.Cu	0.0152 mm	Inner copper
Prepreg	0.0994 mm	3313 RC57%; Dk 4.10
B.Cu	0.0350 mm	Outer nominal 1 oz

Native JLC04081H-3313 basis; copper plus dielectric sum 0.7992 mm. Approximately 10 μ m mask layers are additional. The manufacturer must return the actual finished stackup, plating/etch geometry and thickness.

Feature	Required process
All 137 vias	Nonconductive epoxy fill, planarize and copper cap both ends: IPC-4761 Type VII. 128 x nominal 0.200 mm bores; 9 x nominal 0.304 mm. Follow the via-coordinate CSV. Ordinary tenting is insufficient.
Four J1 plated slots	Keep open: two 0.600 x 1.400 mm and two 0.600 x 1.800 mm. Do not fill connector slots. No NPTH features.
Track / annulus / mask	Minimum track and normal clearance 0.127 mm; smallest via annulus 0.125 mm. U2/U4 local mask expansion 0.050 mm; U2 nominal mask dam 0.100 mm. Preserve supplied mask geometry.

USB: 90 ohms differential $\pm 10\%$

Main B.Cu pair: 0.135 mm width / 0.150 mm gap, about 33.74 mm, over actual filled In2 ground across 0.0994 mm prepreg. The recorded supplier uniform-line model supports these dimensions. Require a representative 81-99 ohm coupon/TDR result and CAM review of launches, vias and nearby ground. Preserve the 0.300 mm coplanar zone spacing and RF keepouts. A coupon does not certify the whole USB channel.

Bare-board acceptance: 100% electrical test against IPC-D-356, AOI, dimensions, filled-via/cap quality and representative microsection/coupon records. Panel rails, tooling and depanel tabs belong outside the final outline and antenna area.

Assembly and first articles

47 fitted SMT references, 27 exact MPN groups. C20, MIC2 and R9 are DNP. Use the fitted-only paste files; do not add paste to test pads. External BT1, M1 and TH1 are installed after reflow.

Stencil and inspection

Starting specification: 80 um laser-cut electropolished stencil, Type 5 SAC305 paste, existing apertures. U4 maximum stencil thickness is 101.6 um. Minimum screened area ratio is 0.994 at 80 um; C13 is 1.118. U2 thermal paste is nine 0.925 mm square windows, 62.86% nominal coverage. Use SPI, profiled two-sided reflow, AOI and first-article hidden-joint X-ray. These checks define the process; production yield is not yet measured.

Closely placed PMIC parts assume +/-0.05 mm placement per part. U4/C1 maximum-body gap after repair is 0.375 mm before that placement allowance. Keep flux, cleaning fluid and debris out of the microphone acoustic port; do not ultrasonically clean it. Solder the USB shell stakes and avoid board bending during depaneling.

External part	Wiring
Jauch LP561836JU+PCM+2 WIRES 50MM 350 mAh minimum protected pack	BT1.1 = positive VBAT BT1.2 = negative GND
VCLP1020B002L motor	M1.1 = 3V_MAIN M1.2 = HAPTIC_NEG
B57540G1103F000, separate 10k NTC	TH1.1 = NTC / TH1.2 = GND Insulate and thermally couple to cell
SWD probe signal	Back-side PCB test pad
SWDIO / SWCLK	TP1 / TP2
nRESET (optional) / VTref sense / GND	TP3 / TP4 (3V_MAIN) / TP5

First-power order

1. Inspect unpowered boards, soldering, shorts and polarity; leave battery disconnected.
2. Apply current-limited 5 V through J1; measure 3V_MAIN before attaching the probe.
3. Program the supplied E1P1 BUCK2-main bench HEX over SWD and verify readback. VTref only senses voltage; do not power the board from it.
4. Capture boot logs, rail/current measurements, audio, BLE/USB, LED/button and motor checks. Keep charging disabled; do not attach an unverified pack.

Limits that remain

Physical hardware tests have not run. The included bench firmware has live PDM/Opus/BLE audio and NAND identification; full offline NAND recording/recovery/backfill, qualified charging and signed updates remain incomplete. Do not deliver these as qualified customer wearables. The board does not fit a 30 x 14 x 8 mm overall shell.

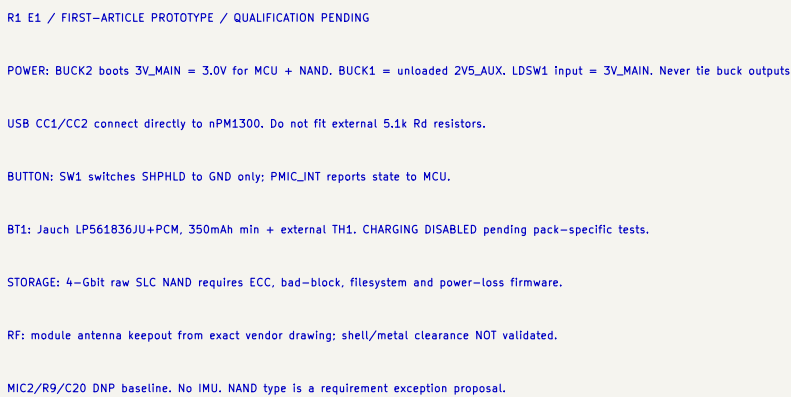
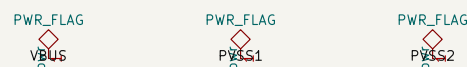
Supply: 26 of 27 MPN groups have sufficient listed distributor stock for 20 boards plus two-board spares. U1 radio modules must be allocated; 25 carrier-delivered modules in BC are consignment candidates, not confirmed usable stock at an assembler.

Fitted placement index

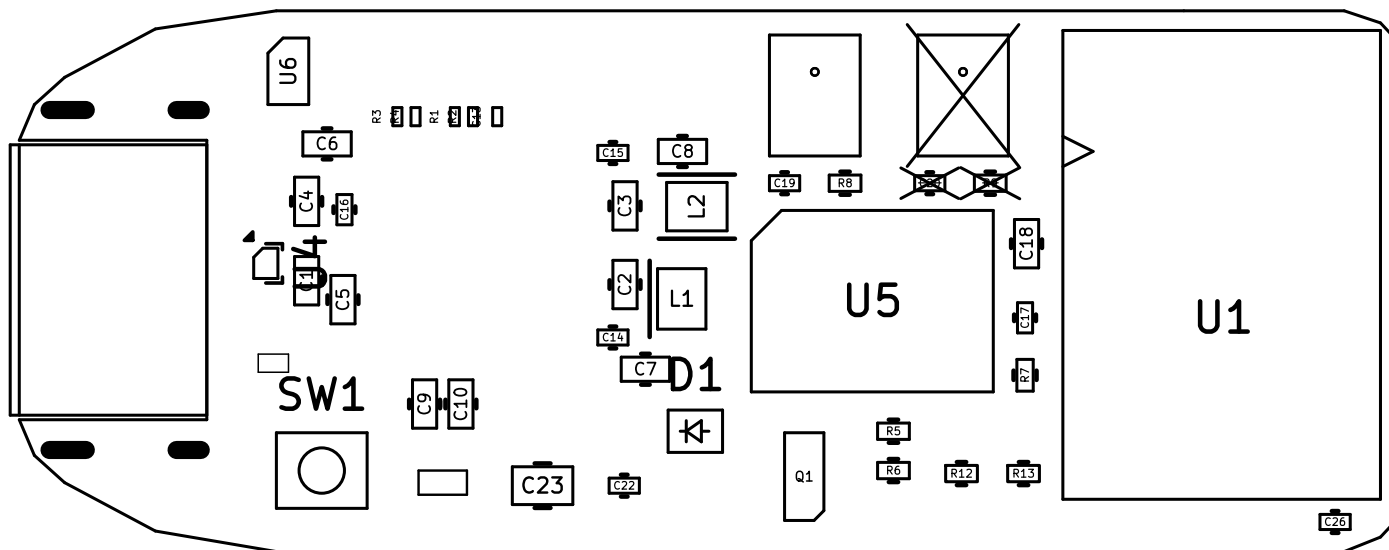
Every machine-fitted reference, including parts whose native F.Fab reference label is hidden. Coordinates and rotation below are the exact exported placement CSV values: millimetres, X right / Y up; T = top, B = bottom. Native editor Y coordinates have the opposite sign. Do not mirror or negate these values a second time for machine import.

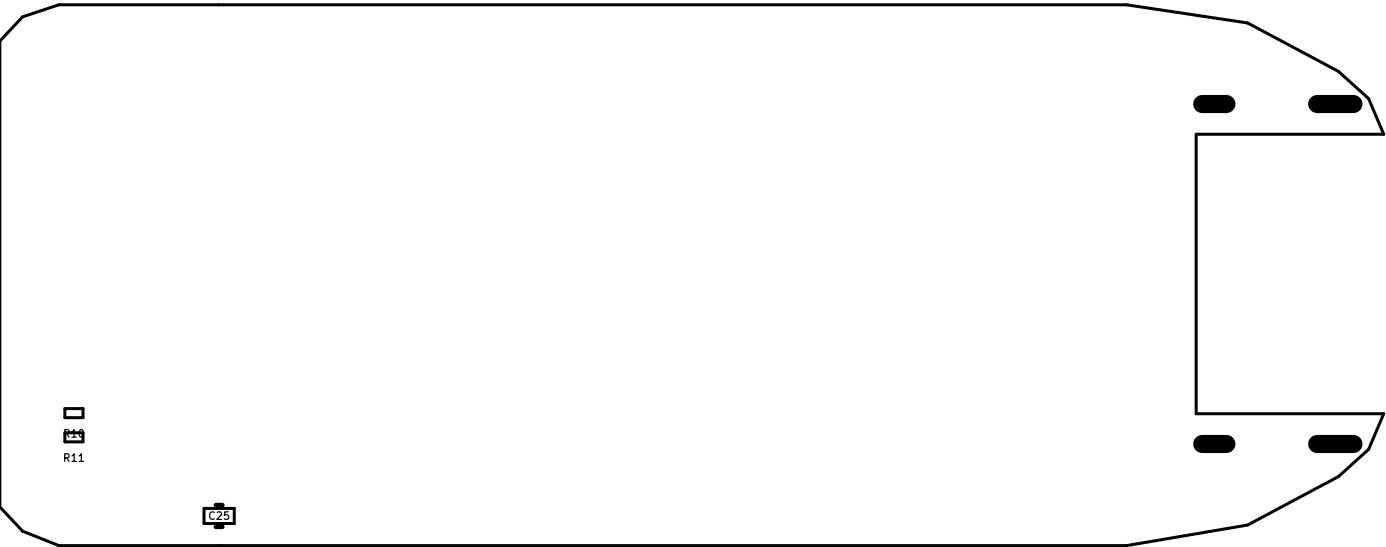
Ref / side	X	Y	Rot.	Ref / side	X	Y	Rot.
C1 / T	29.500	-30.025	-90	L1 / T	41.900	-30.625	90
C10 / T	34.600	-34.100	-90	L2 / T	42.400	-27.575	180
C13 / T	35.800	-24.600	90	LED1 / T	34.000	-36.700	0
C14 / T	39.625	-31.900	0	MIC1 / T	46.300	-23.900	0
C15 / T	39.625	-25.800	0	Q1 / T	45.950	-36.500	180
C16 / T	30.750	-27.675	90	R1 / T	34.400	-24.600	90
C17 / T	53.250	-31.250	90	R10 / B	63.300	-34.600	0
C18 / T	53.300	-28.800	90	R11 / B	63.300	-35.400	0
C19 / T	45.300	-26.800	0	R12 / T	51.150	-36.400	0
C2 / T	40.025	-30.150	-90	R13 / T	53.200	-36.400	0
C22 / T	40.000	-36.800	0	R2 / T	35.000	-24.600	90
C23 / T	37.300	-36.800	0	R3 / T	32.500	-24.600	90
C25 / B	58.500	-38.000	180	R4 / T	33.100	-24.600	90
C26 / T	63.500	-38.000	0	R5 / T	48.900	-35.000	0
C3 / T	40.025	-27.550	90	R6 / T	48.900	-36.300	0
C4 / T	29.500	-27.400	90	R7 / T	53.250	-33.150	90
C5 / T	30.700	-30.650	-90	R8 / T	47.300	-26.800	0
C6 / T	30.175	-25.500	180	SW1 / T	30.000	-36.300	0
C7 / T	40.700	-32.950	180	U1 / T	59.750	-29.500	0
C8 / T	41.925	-25.750	180	U2 / T	35.200	-28.600	180
C9 / T	33.400	-34.100	-90	U4 / T	28.150	-29.450	-90
D1 / T	42.350	-35.000	0	U5 / T	48.200	-30.700	0
D2 / T	28.400	-32.750	0	U6 / T	28.900	-23.100	0
J1 / T	20.000	-30.000	90				

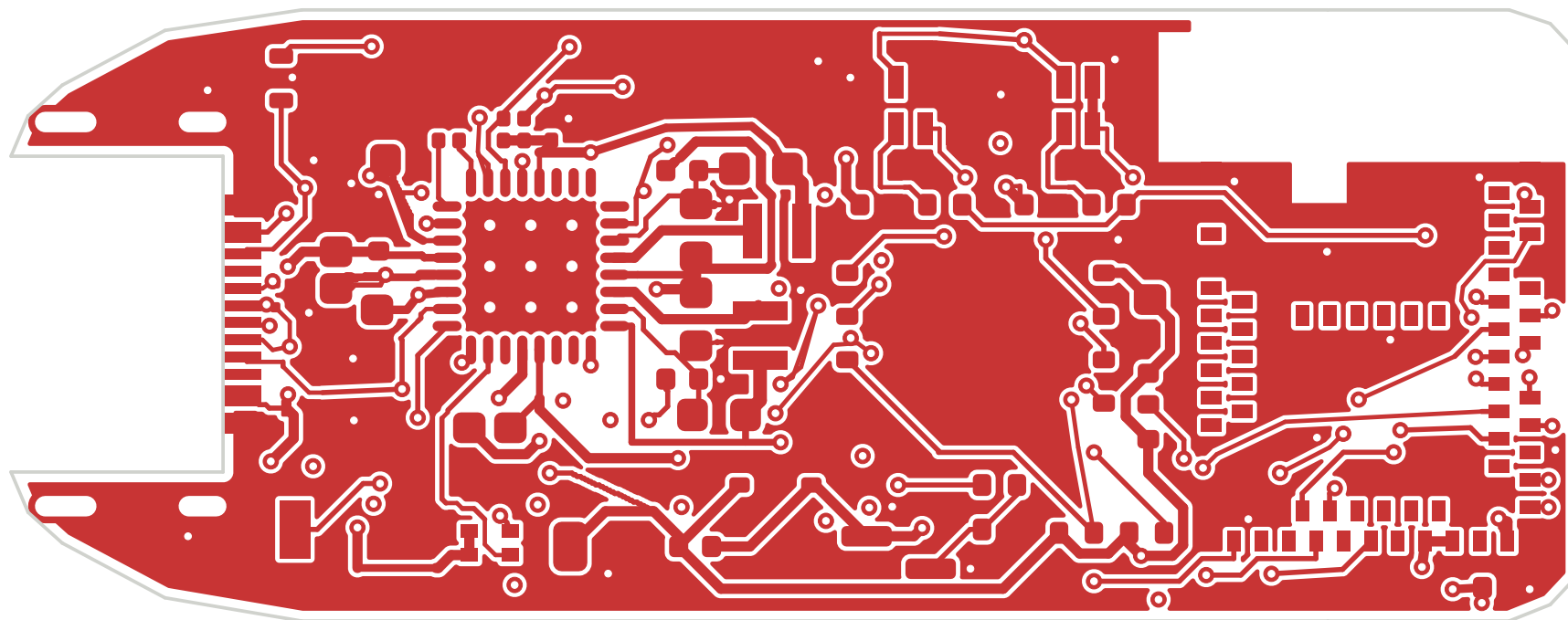
Use the native package drawing, component pin-1 marks and machine-specific rotation convention to validate the first placement. MPNs are in Machine_Assembly_BOM.csv. U2 is at native (35.2,28.6); U4 is at native (28.15,29.45).

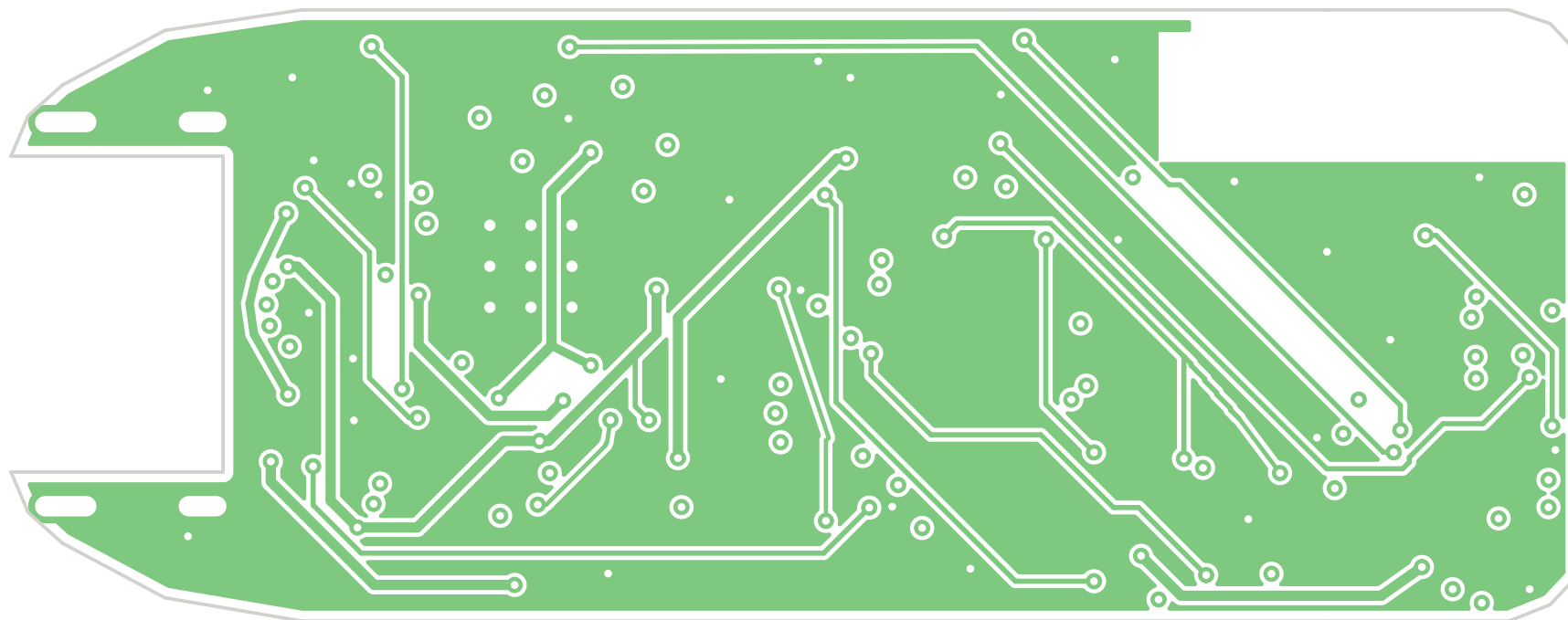


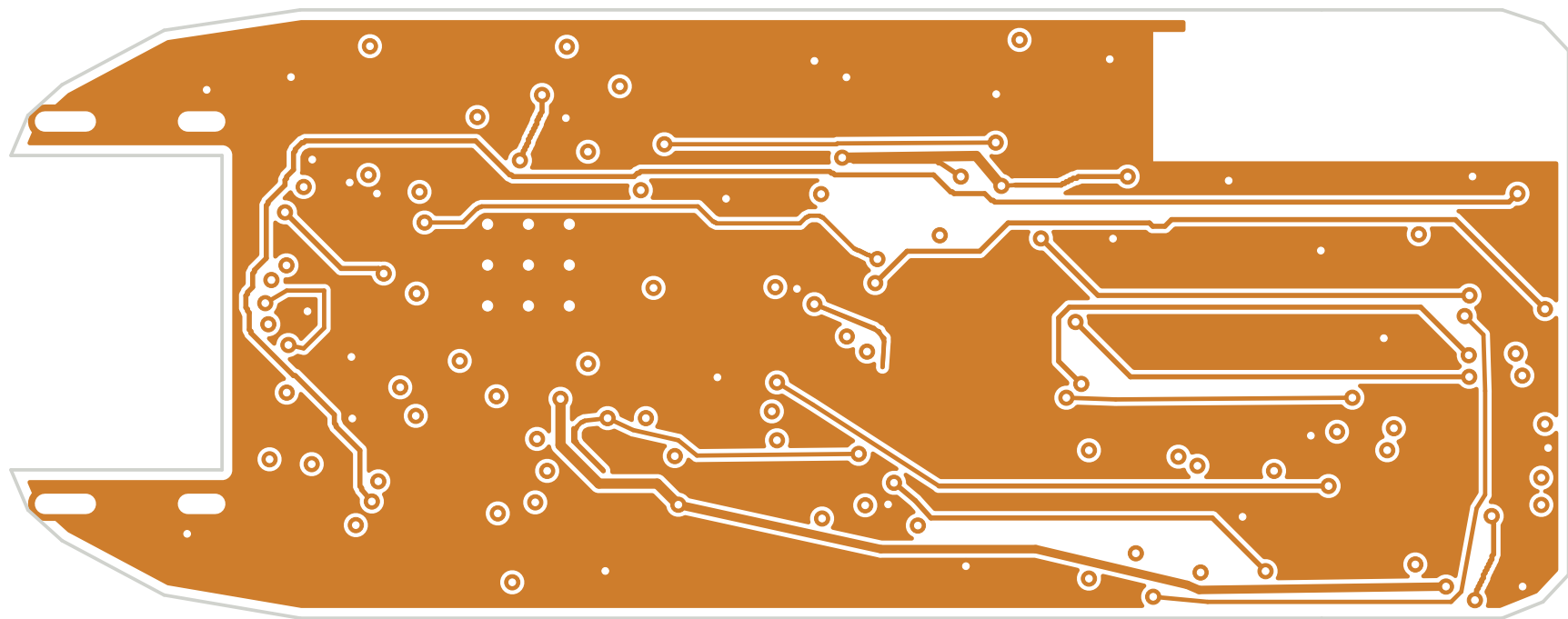
KiCad E.D.A. 10.0.6	Id: 1/1
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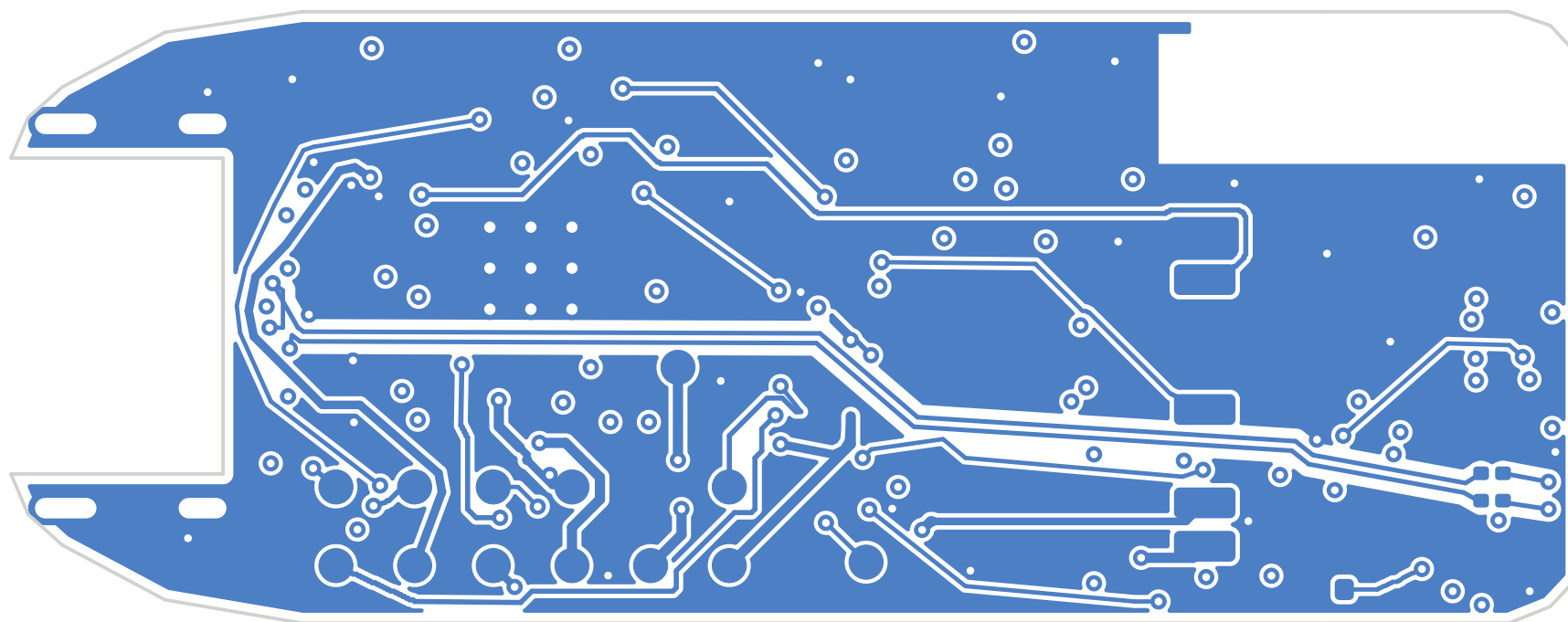


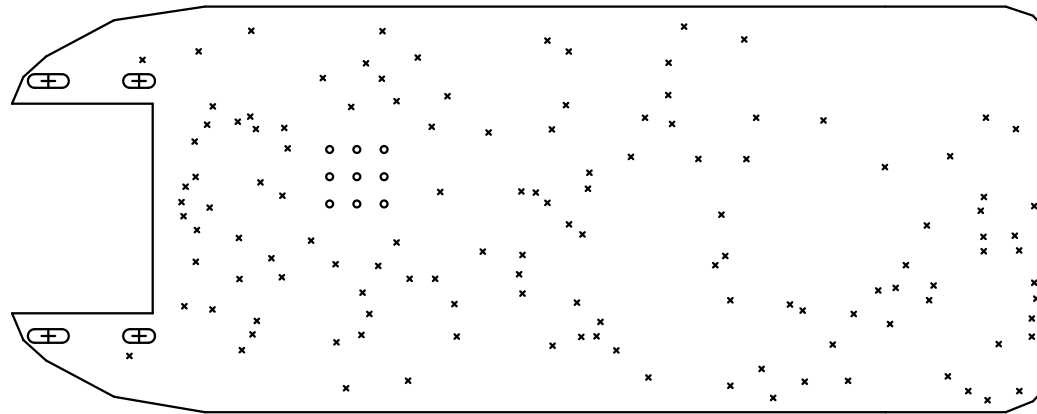












Drill Map:
x 0.200mm / 0.0079" (128 holes)
o 0.304mm / 0.0120" (9 holes)
+ 0.600mm / 0.0236" (0 holes + 4 slots)